

Agent, Affiliate & Admin Panel — Complete Feature Map

Stack: PHP (REST API backend) · MySQL (database) · React.js (frontend SPAs, React Router)

This document covers **every feature** implemented across the three panels, the data each one reads/writes, and exactly how they connect through the Admin console.

Architecture Overview

Public-Facing Panels - Agent Panel (React.js SPA — dollara-agent, React Router) → JWT role=agent → `AgentController.php` / `AgentService.php` - Affiliate Panel (React.js SPA — dollara-affiliate, React Router) → JWT role=affiliate → `AffiliateController.php` / `AffiliateService.php`

Staff Console - Admin Panel (React.js SPA — dollara-admin, React Router) → JWT role=admin → `AgentAdminController.php` / `AgentAdminService.php` and `AffiliateAdminController.php` / `AffiliateAdminService.php`

Backend layering (typical PHP REST setup): `Route (api.php)` → `Controller (validates request, checks JWT role)` → `Service (business logic)` → `Model / Query Builder (PDO or ORM)` → `MySQL`

All four service layers read and write to the same central **MySQL database**.

Trust Boundary: The agent panel can only see its own downline (tree_path prefix match, enforced in the Service layer via a `WHERE tree_path LIKE 'xxx%'` filter). The affiliate panel can only see its own referrals (`WHERE affiliate_id = :id`). The admin panel has no scope restriction — its Controllers skip the ownership filter entirely — it sees everything.

Auth: Stateless JWT (e.g. via `firebase/php-jwt`) issued on login, sent as `Authorization: Bearer <token>`, verified by PHP middleware on every route; the decoded `role` claim (`agent` / `affiliate` / `admin`) drives which Controller/Service layer a request is allowed to reach.

1. AGENT PANEL (dollara-agent — React SPA)

Identity model: `agents` table. Hierarchical tree (super_master → master → agent → sub_agent). Each agent has `tree_path`, `balance`, `credit_reference`, `exposure`, `partnership`, `commission_rate`.

1.1 Authentication & Application

Feature	Page/Route	API Backend	Details
Public Landing	<code>/</code> (React route)	<code>get_program_overview()</code>	Shows programme terms: default partnership %, commission rate, levels, review hours. Pulled from <code>platform_settings.agent_program</code>
Apply	<code>/apply</code>	<code>apply_as_agent()</code>	Form: username, password, name, email, phone, company_name, market_region, expected_volume, experience, notes, parent_code. Creates a <code>pending</code> agent row with no <code>tree_path</code> (invisible to reports). Parent_code optionally resolves an upline
Application Status	<code>/apply/status</code>	<code>application_status(email)</code>	Applicant checks their status by email. Returns status, applied_at, approved_at, rejection_reason
Login	<code>/login</code>	<code>login(username, password)</code>	Validates credential → checks status (pending / info_requested / rejected / closed / suspended / locked all blocked) → mints JWT with <code>role=agent</code>
Change Password	Profile	<code>change_password()</code>	Current password + new password. <code>must_change_password</code> flag cleared on success

1.2 Dashboard (`/dashboard`)

Tile / Section	Data Source	Details
Identity Strip	<code>get_identity()</code>	code, username, name, level, currency, creditReference, balance, exposure, availableCredit, netExposure, partnership%, commissionRate, settledPI, unsettledPI, downlineAgents count, downlinePlayers count
Player Stats Table	<code>_player_stats_rows()</code>	Per-player rows: username, balance, casino bonus, sports bonus, casinoBets/Wins/PL, sportsBets/Wins/PL. Only players active in the date range
New Agents / New Players	<code>get_dashboard()</code>	Count of agents/players created within the range
Active / Last-Hour Players	Game rounds + sport bets	Union of user_ids who placed any bet in the period vs last hour
Total P&L	Casino <code>game_rounds</code> + sports <code>sport_bets</code>	Casino PL = bet_amount – win_amount (excluding sports-category games). Sports PL = exchange profit_loss + aggregator sports rounds
Deposits / Withdrawals	<code>transactions</code> table	Completed deposits/withdrawals sum and count
Revenue Split	Casino vs Sports PL	House-side PL for each product
Top 5 Winning/Losing Players	<code>sport_bets</code> aggregated by user	Member-view (sign flipped): biggest winners and biggest losers
Top 5 Winning/Losing Markets	<code>sport_bets</code> aggregated by market	House-view: best and worst markets for the book
Top 10 Deposit Count	<code>transactions</code>	Players with the most deposit transactions
Top 10 Max Deposit	<code>transactions</code>	Largest individual deposits
Top 10 Withdraw Count	<code>transactions</code>	Players with the most withdrawal transactions

1.3 Clients (Downline Agents) (`/clients`)

Feature	API Function	Details
List Clients	<code>list_clients()</code>	Direct children only (one level). Shows code, username, name, level, creditReference, balance, exposure, availableCredit, partnership, settledPI, unsettledPI, player count, status, locks
Create Client	<code>create_client()</code>	Opens an account below the caller. Fields: username, password, name, level (must be in <code>can_create_below</code>), partnership%, commission_rate%, opening credit (debited from caller's available_credit)
Update Client	<code>update_client()</code>	Modify: betLocked, userLocked, status, partnership%, commissionRate%, password reset (sets <code>must_change_password=True</code>)
Credit Transfer	<code>transfer_credit()</code>	Move credit DOWN to or UP from a direct child (agent or player). Both balances update atomically; <code>credit_reference</code> tracks what was extended

1.4 Players (`/players`)

Feature	API Function	Details
List Players	<code>list_players()</code>	All players under the agent's subtree. Shows username, fullName, availableBalance, balance, currentPI, exposure, Demo/Live type, status
Create Player	<code>create_player()</code>	Opens a <code>users</code> row + <code>wallets</code> row + <code>user_settings.agent_id</code> . NOT the public signup flow (no bonus, no affiliate attribution). Opening credit creates a deposit <code>transactions</code> row
Update Player	<code>update_player()</code>	Change status, password, fullName
Credit Transfer	<code>transfer_credit()</code>	Deposit to or withdraw from a player's wallet. Capped at <code>withdrawable = main_balance – exposure_balance – locked_balance</code>
Settlement	<code>settle()</code>	Record a settlement: amount signed from agent's side. Moves <code>unsettled_pl</code> → <code>settled_pl</code> on the agent

1.5 Sport Analysis (`/sport-analysis`)

Feature	API Function	Details
Live Book	<code>get_sport_analysis()</code>	Every event with open bets under this agent, grouped by sport. No date filter — shows current risk
Event Drill-Down	<code>get_event_book(event_id)</code>	All open bets on a single event: player, market, selection, side, odds, stake, liability, exposure, P&L, status
Per-Market Breakdown	Inside sport analysis	Markets within each event: exposure (what the book stands to lose), maxProfit

1.6 Reports (8 reports)

All reports share: date range filter, sport/marketType/event/agentName/status filters, credit summary strip, grand total row, CSV export.

Report	Route	API Function	Groups By	Key Columns
P&L by Market	<code>/reports/pl-market</code>	<code>pl_by_market()</code>	Event + Market	Bets, Turnover, MemberWin/Comm/PL, AgentWin/Comm/PL, UplinePL
P&L by Agent	<code>/reports/pl-agent</code>	<code>pl_by_agent()</code>	Agent	Same P&L split, each row uses the ROW agent's own partnership%
Event P&L	<code>/reports/event-pl</code>	<code>event_pl()</code>	Event	Players, Bets, Turnover, Member/Agent/Upline PL
Bet List	<code>/reports/bet-list</code>	<code>bet_list()</code>	Individual bets	Player, event, market, selection, side, odds, stake, liability, P&L, status. Paginated
Transfer Statement	<code>/reports/transfer-statement</code>	<code>transfer_statement()</code>	Credit ledger	Date, counterparty, direction, signed amount, balance after, remark
Settlement	<code>/reports/settlement</code>	<code>settlement_report()</code>	Settlements	Date, counterparty, amount, PL before/after, period, note
Transactions	<code>/reports/transactions</code>	<code>transactions_report()</code>	Deposits & Withdrawals	Player, type, amount, status, method, reference
Real Revenue	<code>/reports/real-revenue</code>	<code>real_revenue()</code>	Per-agent	Deposits, Withdrawals, Net Cash, Sports PL, Casino PL, Gross Revenue, Real Revenue

P&L Sign Convention: Stored as house-positive. `_split_pl()` flips the sign once: $\text{memberWin} = -\text{house_pl}$. Partnership% determines what the viewing agent keeps vs what flows to upline.

1.7 Profile

Feature	Details
View Identity	Name, code, level, currency, timezone, all financial figures
Change Password	Current + new, clears <code>must_change_password</code>

2. AFFILIATE PANEL (dollara-affiliate — React SPA)

Identity model: `affiliates` table. Tree structure via `parent_id`. Commission model: Revenue Share, CPA, or Hybrid. Has tiers (bronze/silver/gold/platinum/diamond), 2FA, KYC, and API keys.

2.1 Authentication & Application

Feature	Route	API Backend	Details
Public Landing	/ (React route)	<code>get_program_overview()</code>	Shows commission structure, tiers, default rates, cookie duration
Apply	<code>/apply</code>	<code>apply_as_affiliate()</code>	Form: full_name, email, password, phone, company_name, website, marketing_channels, monthly_traffic, experience, referral_source, parent_affiliate_code. Creates <code>pending</code> affiliate row
Application Status	Built into apply	<code>application_status(email)</code>	Same pattern as agent: check by email
Login	<code>/login</code>	<code>login(email, password)</code>	Email-based auth. Supports 2FA flow: returns <code>challengeToken</code> if 2FA is enabled
2FA Verify	Login flow	<code>verify_two_factor()</code>	TOTP code verification against the challenge token
Forgot Password	<code>/login</code>	<code>request_password_reset()</code>	Issues a challenge token
Reset Password	Via token	<code>reset_password()</code>	Consumes the challenge and sets new password
Onboarding	<code>/onboarding</code>	<code>get_onboarding()</code> , <code>complete_onboarding()</code>	Post-approval checklist: profile completion, KYC document upload, payout method setup. Must complete before accessing dashboard

2.2 Dashboard (`/dashboard`)

Tile / Section	API Function	Details
KPI Cards	<code>get_dashboard()</code>	Clicks, Signups, FTDs (first-time depositors), Active players, Commission earned, Conversion rate, EPC (earnings per click), Revenue
Performance Chart	<code>get_dashboard_chart()</code>	Time-series data: daily clicks, signups, FTDs, commission. Breakdown by day/week/month
Activity Feed	<code>get_activity()</code>	Recent events: new signups, FTDs, commission entries, payouts
Top Links Table	Dashboard component	Best-performing tracking links by clicks/conversions
Commission Funnel	Dashboard component	Click → Signup → FTD → Revenue funnel visualization

2.3 Links Management (`/links`)

Feature	API Function	Details
List Links	<code>list_links()</code>	All tracking links: name, code, sub_id, target_path, tracking URL, click/signup/FTD stats
Create Link	<code>create_link()</code>	Name, sub_id (optional campaign tag), target_path. Auto-generates a unique link code
Update Link	<code>update_link()</code>	Change name, sub_id, target_path, status
Delete Link	<code>delete_link()</code>	Soft-delete the link
Link Directory	<code>list_creatives()</code> , <code>list_landing_pages()</code>	Browse available banners/creatives and landing pages to link to
Click Tracking	<code>record_click()</code>	Server-side: records IP, user_agent, referrer, geo, sets cookie. Deduplicates by IP+affiliate within a window

2.4 Referrals (`/referrals`)

Feature	API Function	Details
List Referrals	<code>list_referrals()</code>	All referred users: username, status (click/signup/ftd/active/churned), signup_at, first_deposit_at, lifetime_commission, link_name. Filterable by status, search, date
Referral Detail	<code>get_referral_detail()</code>	Deep view: activity timeline, deposit/wager history, commission entries generated, qualified status
Attribution	<code>attribute_signup()</code>	Called by the platform during user registration. Resolves <code>ref</code> code → affiliate, creates <code>affiliate_referrals</code> row, runs fraud checks
Deposit Recording	<code>record_deposit()</code>	Called on every deposit. If referral has no FTD, marks it. Computes rev-share commission or CPA on qualifying deposit

2.5 Network (Sub-Affiliates) (`/network`)

Feature	API Function	Details
Sub-Affiliate Tree	<code>get_network()</code>	Recruited partners, one level deep. Per child: code, name, status, tier, clicks, signups, FTDs, their commission earned, override rate, override earned by you. Plus grandchildren summary
Override Summary	Inside network	Total override commission earned, average override rate, network signups
Create Invite	<code>create_invite()</code>	Generates invite URL with parent_affiliate_code and override_rate (capped at program default)

2.6 Finance — Earnings (`/finance/earnings`)

Feature	API Function	Details
Commission Ledger	<code>list_earnings()</code>	Every commission entry: entry_type (rev_share / cpa / override), period, base_kind (GGR/NGR/deposits/wagers), base_amount, rate%, amount, currency, status (pending/approved/paid/clawed_back), source_affiliate
Summary Cards	Inside earnings	Pending, Approved, Paid, Clawed Back totals. Commission type and rate
CSV Export	<code>earnings_csv_rows()</code>	Full statement export: Entry ID, Period, Type, Base, Amount, Rate, Status, Source

2.7 Finance — Payouts (`/finance/payouts`)

Feature	API Function	Details
Payout Dashboard	<code>get_payouts()</code>	Available balance (approved & unclaimed), pending commission, payout threshold, payout cycle (weekly/monthly), next payout date, has open request
Request Payout	<code>request_payout()</code>	Creates a payout request against available balance. Locks the commission entries to the payout. Must have a primary payout method, balance $\geq$ threshold, no existing open request
Payout History	Inside payouts	All payouts: amount, status (requested/approved/paid/rejected), method, reference, created/paid dates, entry count
Payout Methods	<code>list_payout_methods()</code> , <code>upsert_payout_method()</code>	Types: bank_transfer, UPI, crypto, ewallet. Fields: method_type, account details JSON, is_primary

2.8 Reports (`/reports`)

Feature	API Function	Details
Performance Reports	<code>get_reports()</code>	Breakdown by: link, date, sub_id, country. Columns: clicks, unique clicks, signups, FTDs, deposits, wagers, commission, conversion rate, EPC. Supports CSV export
Date Range Filter	Context-level	7d / 30d / 90d presets or custom range. Shared via <code>AffiliateContext</code>

2.9 Notifications (`/notifications`)

Feature	API Function	Details
List Notifications	<code>list_notifications()</code>	All notifications: type, title, message, read/unread, created_at. Unread badge in header
Mark Read/Unread	<code>mark_notification()</code>	Toggle individual notification
Mark All Read	<code>mark_all_notifications_read()</code>	Bulk read
Clear All	<code>clear_notifications()</code>	Delete all notifications

2.10 Support (`/support`)

Feature	API Function	Details
List Tickets	<code>list_tickets()</code>	Support tickets: subject, category (technical/payment/commission/other), status (open/in_progress/resolved/closed), message count, created_at
Create Ticket	<code>create_ticket()</code>	Subject, message, category, optional attachment
View Thread	<code>get_ticket()</code>	Full message thread with timestamps and sender type
Reply	<code>add_ticket_message()</code>	Append a message to the thread

2.11 Settings

Feature	Route	API Function	Details
Profile	/settings/profile	get_profile() , update_profile()	Update name, company, phone, website, marketing_channels, bio
Password	/settings/profile	change_password()	Current + new password
Two-Factor Auth	/settings/profile	start_two_factor() , enable_two_factor() , disable_two_factor()	TOTP setup with QR code, enable with code verification, disable requires password + code
KYC Documents	/settings/profile	upload_kyc_document()	Upload identity, address, or company documents. Types: id_front, id_back, proof_of_address, company_registration
API Keys	/settings/api	list_api_keys() , create_api_key() , rotate_api_key() , revoke_api_key()	Generate API keys for programmatic access. Rotation with 24hr grace period. Max keys capped

3. ADMIN PANEL — AGENT MANAGEMENT (dollara-admin/src/pages/agents/)

Backend: `AgentAdminController.php` / `AgentAdminService.php` — No scope restriction. Staff sees every agent.

3.1 Agent List (/agents)

Feature	API Function	Details
List All Agents	list_agents()	Every approved agent (applications excluded). Filterable by status, level, search (name/username/code/email). Shows: code, username, name, level, depth, tree_path, parent, status, partnership%, commission_rate%, credit_reference, balance, exposure, available_credit, settled_pl, unsettled_pl, player count, downline count, locks, join date, last login
Summary Cards	_list_summary()	Active count, Suspended count, Total balance, Total exposure, Total unsettled P&L, Pending applications count
Level Filter	From <code>Agent.Level.choices</code>	super_master, master, agent, sub_agent

3.2 Agent Detail (/agents/[id])

Tab / Section	API Function	Details
Identity	get_agent_detail()	All fields from the agents table: code, username, name, level, depth, tree_path, parent, status, partnership, commission_rate, commission_type, credit_reference, balance, exposure, available_credit, settled_pl, unsettled_pl, total_commission, pending_commission, locks, must_change_password, join date, last login IP
Stats	Inside detail	downline_agents count, downline_players count, direct_children count, downline_balance sum, net_exposure
Downline List	Inside detail	Direct children with their balance, exposure, partnership, player count
Players List	_direct_players()	Players attached to this agent: username, name, status, balance, exposure, join date
Transfer History	Inside detail	Last 100 transfers: counterparty, direction, amount, balance_after, remark, is_platform (injection)
Settlement History	Inside detail	Last 100 settlements: counterparty, amount, PL before/after, period, note
Audit Log	Inside detail	Last 100 audit entries: actor, action, target, metadata, IP, timestamp

3.3 Agent Actions (from detail page)

Action	API Function	Details
Update Terms	<code>update_agent()</code>	Change: partnership%, commissionRate%, betLocked, userLocked, name, contactEmail, contactPhone, level, parentAgentId (reparent — with cycle detection and subtree path restamping)
Set Status	<code>set_agent_status()</code>	Move between: active, suspended, locked, closed. <code>is_active</code> follows status automatically
Reset Password	<code>reset_password()</code>	Sets new password + <code>must_change_password=True</code>
Adjust Credit	<code>adjust_credit()</code>	Platform injection: positive funds, negative claws back. Self-referencing transfer row so it appears on the agent's Transfer Statement. Capped at available_credit for negative
Delete Agent	<code>delete_agent()</code>	Only if: no downline, no players, zero balance, zero exposure

3.4 Applications (`/agents/applications`)

Feature	API Function	Details
List Applications	<code>list_applications()</code>	All pending/info_requested applications. Shows: applicant details (name, email, phone, company, market_region, expected_volume, experience, notes), requested parent code, resolved parent, status
Approve	<code>approve_application()</code>	Sets: parent (or root), level, partnership%, commission_rate%, opening_credit. Attaches to tree (writes tree_path + depth). Debits parent's balance for credit, or injects platform credit for roots
Reject	<code>decide_application(decision='rejected')</code>	Requires a reason (shown to applicant)
Request Info	<code>decide_application(decision='info_requested')</code>	Requires a message (shown to applicant)

3.5 Credit Management (`/agents/credit`)

Feature	API Function	Details
List Transfers	<code>list_transfers()</code>	All credit transfers across all agents. Filterable by agent, direction, date range. Shows: agent, counterparty, is_platform flag, direction, amount, balance_after, remark
List Settlements	<code>list_settlements()</code>	All settlements. Filterable by agent, date range. Summary: total settled amount

3.6 Audit Log (`/agents/audit`)

Feature	API Function	Details
Global Audit	<code>list_audit_log()</code>	Every agent action (panel + admin). Filterable by agent, action keyword, date range. Actions logged: auth.login, auth.password_changed, client.created, client.updated, player.created, player.updated, credit.down, credit.up, settlement.recorded, application.submitted, admin.application.approved/rejected, admin.agent.updated, admin.credit.adjusted, admin.settings.updated, etc.

3.7 Programme Settings (`/agents/settings`)

Feature	API Function	Details
View Settings	<code>get_settings()</code>	default_level, default_partnership%, default_commission_rate%, default_opening_credit, min/max_partnership, review_hours, currency
Update Settings	<code>update_settings()</code>	Merge-saves over stored JSON in <code>platform_settings.agent_program</code> . Audit logged with before/after

4. ADMIN PANEL — AFFILIATE MANAGEMENT (`dollara-admin/src/pages/affiliates/`)

Backend: `AffiliateAdminController.php` / `AffiliateAdminService.php`

4.1 Affiliate List (`/affiliates`)

Feature	API Function	Details
List All Affiliates	<code>list_affiliates()</code>	Filterable by status, tier, commission_type, search. Shows: code, name, email, company, status, tier, commission_type, commission_rate, cpa_amount, clicks, signups, FTDs, total/pending/paid commission, lifetime revenue, balance, parent, joined_at
Summary Cards	Inside list	Active/suspended count, total clicks, total signups, total FTDs, total commission paid, pending payouts

4.2 Affiliate Detail (`/affiliates/[id]`)

Tab / Section	API Function	Details
Full Profile	<code>get_affiliate_detail()</code>	All affiliate fields + stats (total referrals, active referrals, FTD count, conversion rate, total commission, paid, pending, available balance, lifetime revenue)
Links	Inside detail	All tracking links with performance stats
Referrals	Inside detail	Recent referrals with status and commission
Commission Ledger	Inside detail	Recent commission entries
Payouts	Inside detail	Payout history
KYC Documents	Inside detail	Uploaded documents with review status
API Keys	Inside detail	Active/revoked keys
Sub-Affiliates	Inside detail	Network tree
Audit Log	Inside detail	All actions on this affiliate

4.3 Affiliate Actions

Action	API Function	Details
Update Terms	<code>update_affiliate()</code>	Change: name, commission_type (rev_share/cpa/hybrid), commission_rate%, cpa_amount, override_rate%, payout_threshold, tier, parent (reparent with cycle detection), contactEmail, contactPhone
Set Status	<code>set_affiliate_status()</code>	active, suspended, closed
Delete	<code>delete_affiliate()</code>	Removes affiliate + cascades referrals, links, ledger
Review KYC	<code>review_kyc_document()</code>	Approve or reject uploaded documents. Status + reviewer note
Revoke API Key	<code>revoke_affiliate_key()</code>	Deactivate a specific API key

4.4 Applications (`/affiliates/applications`)

Feature	API Function	Details
List Applications	<code>list_applications()</code>	Pending/info_requested affiliates
Approve	<code>approve_application()</code>	Sets: commission_type, commission_rate, cpa_amount, tier, parent, status → active
Reject	<code>reject_application()</code>	With reason
Request Info	<code>request_application_info()</code>	With message

4.5 Payouts (`/affiliates/payouts`)

Feature	API Function	Details
List Payout Requests	<code>list_payout_requests()</code>	All pending/approved payouts. Summary: pending count, total requested amount
Approve Payout	<code>approve_payout()</code>	Moves status → approved
Mark Paid	<code>mark_payout_paid()</code>	Moves status → paid, records reference number, updates all linked commission entries to "paid"
Reject Payout	<code>reject_payout()</code>	Returns money to available balance, unlinks commission entries from the payout
Bulk Action	<code>bulk_payout_action()</code>	Approve/reject/pay multiple payouts at once

4.6 Commission Management

Feature	API Function	Details
Approve Entry	<code>approve_ledger_entry()</code>	Move commission entry from pending → approved
Clawback	<code>clawback_ledger_entry()</code>	Reverse a commission entry with reason. Creates a negative offsetting entry
Trigger Run	<code>trigger_commission_run()</code>	Manually trigger commission calculation for a date range, optionally for a single affiliate
List Runs	<code>list_commission_runs()</code>	History of commission calculation batches

4.7 Fraud Detection

Feature	API Function	Details
List Fraud Flags	<code>list_fraud_flags()</code>	Filterable by status (open/resolved/dismissed), risk level. Auto-detected: duplicate IPs, same email domain clusters, rapid signups, self-referral attempts
Resolve Flag	<code>resolve_fraud_flag()</code>	Mark as: confirmed (freeze affiliate), dismissed (false positive), or investigating

4.8 Audit & Settings

Feature	API Function	Details
Audit Log	<code>list_audit_log()</code>	All affiliate-related actions, filterable by affiliate, date range
Programme Settings	<code>get_settings()</code> , <code>update_settings()</code>	default_commission_rate%, default_cpa_amount, default_override_rate%, min_payout_threshold, cookie_duration_days, payout_cycle, auto_approve_commissions, fraud_check_enabled

5. KEY CONNECTIONS: How the Three Panels Interrelate

5.1 Agent ↔ Admin Connection Map

Flow: Agent Panel Actions → Admin Response

- Agent applies via `/apply` → pending agent row created → Admin reviews in `/agents/applications`
- Admin approves → sets `tree_path` + credit → Agent logs in and works
- Agent logs in / works → all actions audit-logged → Admin monitors in `/agents` + `/agents/audit`
- Agent creates sub-agents → child agent rows created → Visible in Agent Detail downline
- Agent transfers credit → `agent_transfers` rows → Visible in `/agents/credit` transfers
- Agent creates players → `user_settings.agent_id` set → Visible in Agent Detail players tab

Agent Feature	What Admin Sees/Controls
Agent applies	Application appears in Admin <code>/agents/applications</code> . Admin can approve (set level, parent, partnership%, credit), reject, or request more info
Agent logs in	Login event in audit log. Last login time + IP on detail page
Agent creates sub-agent	New agent row appears in Admin agent list. Credit deducted from parent visible in transfers
Agent creates player	Player appears in Admin agent detail “Players” tab. Opening credit recorded as <code>agent_transfer</code> + transaction
Agent transfers credit	Transfer row visible in Admin <code>/agents/credit</code> . Both balances tracked
Agent makes settlement	Settlement row visible in Admin agent detail “Settlements” tab
Agent views reports	No admin visibility (read-only). But the underlying bets/transactions are visible
Agent’s book has exposure	Exposure shown in Admin agent list + detail. Net exposure includes downline
Agent is at risk	Admin can: suspend, lock, close, lock bets, reset password, claw back credit
Programme terms change	Admin updates in <code>/agents/settings</code> → affects new applications and the public landing page

5.2 Affiliate ↔ Admin Connection Map

Flow: Affiliate Panel Actions → Admin Response

- Affiliate applies via `/apply` → pending affiliate row → Admin reviews in `/affiliates/applications`
- Affiliate completes onboarding/KYC → `kyc_documents` created → Admin reviews KYC in affiliate detail
- Affiliate creates tracking links → `affiliate_links` rows → Links visible in affiliate detail
- Users click links → sign up → deposit → clicks + referrals + fraud checks → Referrals + fraud flags visible
- Affiliate earns commission → `commission_ledger` entries → Commission ledger, approve/clawback
- Affiliate requests payout → `affiliate_payouts` row → Payout queue in `/affiliates/payouts`
- Affiliate opens support ticket → `support_tickets` row → Ticket visible in support system

Affiliate Feature	What Admin Sees/Controls
Affiliate applies	Application in Admin <code>/affiliates/applications</code> . Admin sets: commission_type, rate, CPA amount, tier, parent
KYC uploaded	Appears in detail page. Admin approves/rejects each document
Links created	Visible in detail page. Admin can see performance stats
Click → Signup → FTD	Referrals visible in detail. Fraud checks auto-run: duplicate IP, email domain, rapid signups. Flags appear in fraud queue
Commission earned	Entries in ledger. Admin can: approve (pending → approved), clawback (with reason, creates negative offset), trigger commission run
Payout requested	Appears in <code>/affiliates/payouts</code> queue. Admin can: approve → mark paid (with reference), reject (returns balance), bulk action
Support ticket	Visible in support system. Admin can reply and change status
API key created	Visible in detail page. Admin can revoke
Sub-affiliate recruited	Network visible in detail. Override commission entries trackable
Fraud detected	Auto-flags in fraud queue. Admin resolves: confirmed (freeze), dismissed, investigating
Programme terms change	Admin updates in <code>/affiliates/settings</code> → affects defaults for new affiliates

5.3 Agent vs Affiliate — Key Differences

Aspect	Agent Panel	Affiliate Panel
Identity	Username-based login	Email-based login
Auth Security	Password only	Password + optional TOTP 2FA
Tree Model	4-level hierarchy (super_master → master → agent → sub_agent)	Flat + 1-level sub-affiliates (parent → child)
Revenue Model	Partnership % split of P&L	Revenue share %, CPA per FTD, or Hybrid
Money Flow	Credit system: agent balance ↔ player wallet	Commission ledger → payout requests
Player Relationship	Direct: agent creates and manages players	Indirect: affiliate drives traffic, platform manages users
Scope	Tree-path prefix match (can only see downline)	Own referrals only
Reports	8 operational reports (P&L, bets, transfers, revenue)	Performance reports (clicks, signups, FTDs, commission)
Settlement	Agent settles P&L with downline manually	Commission auto-calculated, payout through request workflow
Risk	Real-time exposure tracking, sport analysis, open book	Fraud detection (duplicate IPs, self-referral, rapid signup)
KYC	Not implemented	Full document upload + admin review
API Access	Not implemented	API keys with rotation and grace periods
Support	Not implemented	Ticket system with threaded messages
Notifications	Not implemented	Full notification system with badge count
Onboarding	Login after approval	Post-approval checklist (profile + KYC + payout method)

6. Database Tables Summary

Agent System Tables

- `agents` — The agent accounts (also holds applications as status=pending)
- `agent_transfers` — Credit movement ledger (agent↔agent, agent↔player, platform injection)
- `agent_settlements` — Settlement records
- `agent_audit_logs` — Every action logged
- `sport_bets` — Exchange bets (the book)
- `sport_events` , `sport_markets` — Event/market reference data
- `user_settings.agent_id` — Player → Agent attribution
- `platform_settings` (key= `agent_program`) — Programme configuration

Affiliate System Tables

- `affiliates` — Affiliate accounts + applications
- `affiliate_links` — Tracking links
- `affiliate_clicks` — Click events
- `affiliate_referrals` — Attributed signups
- `affiliate_commission_ledger` — Every commission entry
- `affiliate_payouts` — Payout requests
- `affiliate_payout_methods` — Bank/UPI/crypto details
- `affiliate_api_keys` — Programmatic access
- `affiliate_kyc_documents` — Uploaded identity docs
- `affiliate_support_tickets` + `affiliate_ticket_messages` — Support
- `affiliate_fraud_flags` — Auto-detected fraud
- `affiliate_login_challenges` — 2FA and password reset tokens
- `affiliate_creatives` — Marketing banners/assets
- `affiliate_audit_logs` — Every action logged
- `notifications` — In-app notifications
- `platform_settings` (key= `affiliate_program`) — Programme configuration

Shared Tables (used by both)

- `users` — Player accounts
- `wallets` — Player balances
- `transactions` — Deposits and withdrawals
- `game_rounds` — Casino play (aggregator callbacks)